

techniques developed for its proof, the various ways in which minors are handled or constructed. Most of these have not even been touched upon here, yet they seem set to influence the development of graph theory for many years to come.

Exercises

- 1.⁻ Let $\leq$ be a quasi-ordering on a set X . Call two elements $x, y \in X$ *equivalent* if both $x \leq y$ and $y \leq x$. Show that this is indeed an equivalence relation on X , and that $\leq$ induces a partial ordering on the set of equivalence classes.
2. Let $(A, \leq)$ be a quasi-ordering. For subsets $X \subseteq A$ write

$$\text{Forb}_{\leq}(X) := \{a \in A \mid a \not\geq x \text{ for all } x \in X\}.$$

Show that $\leq$ is a well-quasi-ordering on A if and only if every subset $B \subseteq A$ that is closed under $\leq$ (i.e. such that $x \leq y \in B \Rightarrow x \in B$) can be written as $B = \text{Forb}_{\leq}(X)$ with finite X .

3. Prove Proposition 12.1.1 and Corollary 12.1.2 directly, without using Ramsey's theorem.
- 4.⁻ Show that the relation $\leq$ between rooted trees defined in the text is indeed a quasi-ordering.
5. Show that the finite trees are not well-quasi-ordered by the subgraph relation.
6. The last step of the proof of Kruskal's theorem considers a 'topological' embedding of T_m in T_n that maps the root of T_m to the root of T_n . Suppose we assume inductively that the trees of A_m are embedded in the trees of A_n in the same way, with roots mapped to roots. We thus seem to obtain a proof that the finite rooted trees are well-quasi-ordered by the subgraph relation, even with roots mapped to roots. Where is the error?
7. Extend Kruskal's theorem to trees whose vertices are labelled from a well-quasi-ordered set. The tree embedding is defined as before but in addition respects the ordering of the labels.
8. Are the connected finite graphs well-quasi-ordered by contraction alone (i.e. by taking minors without deleting edges or vertices)?
- 9.⁺ Relax the minor relation by not insisting that branch sets be connected. Show that the finite graphs are well-quasi-ordered by this relation.
- 10.⁺ Show that the finite graphs are not well-quasi-ordered by the topological minor relation.
- 11.⁺ Given $k \in \mathbb{N}$, is the class $\{G \mid G \not\geq P^k\}$ well-quasi-ordered by the subgraph relation?

- 12.⁻ Let G be a graph, T a tree, and $\mathcal{V} = (V_t)_{t \in T}$ a family of subsets of $V(G)$. Show that $(T, \mathcal{V})$ is a tree-decomposition of G if and only if
- (i) for every $v \in V(G)$ the set $T_v := \{t \mid v \in V_t\}$ is connected in T ;
 - (ii) $T_u \cap T_v \neq \emptyset$ for every edge uv of G .
- 13.⁻ Consider a tree-decomposition of a graph G in which some parts contain other parts. Modify it into a tree-decomposition whose parts are the $\subseteq$ -maximal parts of the first decomposition. How does the new tree arise from the old?
14. Let G be a graph, T a set, and $(V_t)_{t \in T}$ a family of subsets of $V(G)$ satisfying (T1) and (T2) from the definition of a tree-decomposition. Show that there exists a tree on T that makes (T3) true if and only if there exists an enumeration $t_1, \dots, t_n$ of T such that for every $k = 2, \dots, n$ there is a $j < k$ satisfying $V_{t_k} \cap \bigcup_{i < k} V_{t_i} \subseteq V_{t_j}$.
- (The new condition tends to be more convenient to check than (T3). It can help, for example, with the construction of a tree-decomposition into a given set of parts.)
15. Prove the following converse of Lemma 12.3.1: if $(T, \mathcal{V})$ satisfies condition (T1) and the statement of the lemma, then $(T, \mathcal{V})$ is a tree-decomposition of G .
16. Recall that two separations $\{U_1, U_2\}$ and $\{W_1, W_2\}$ of G are *nested* if we can choose $i, j \in \{1, 2\}$ so that $U_i \subseteq W_j$ and $U_{3-i} \supseteq W_{3-j}$.
- (i) Show that the separations $S_e := \{U_1, U_2\}$ in Lemma 12.3.1 are pairwise nested (for different choices of the edge $e = t_1 t_2 \in T$).
 - (ii)⁺ Conversely, show that given a set N of nested separations of G there is a tree-decomposition $(T, \mathcal{V})$ of G such that $N = \{S_e \mid e \in E(T)\}$.
- 17.⁺ Prove Theorem 12.3.7 for $k = 3$. Specifically, prove Tutte's theorem that every 2-connected graph has a tree-decomposition of adhesion 2 whose torsos are each either 3-connected or a cycle. Conversely, show that every graph with such a tree-decomposition is 2-connected.
- (Hint. Try the tree-decomposition defined, as in Exercise 16 (ii), by the set of all separations of order 2 that are nested with all other such separations.)
18. Describe the tree-decomposition of a contraction minor H of G which a given tree-decomposition of G induces as in Lemma 12.3.3, in terms subtrees of T (as in Exercise 12).
- 19.⁻ Show that any graph with a simplicial tree-decomposition into k -colourable parts is itself k -colourable.
20. Let $\mathcal{H}$ be a set of graphs, and let G be constructed recursively from elements of $\mathcal{H}$ by pasting along complete subgraphs. Show that G has a simplicial tree-decomposition into elements of $\mathcal{H}$.

21. Use the previous exercise to show that G has no K^5 minor if and only if G has a tree-decomposition in which every torso is either planar or a copy of the Wagner graph W (Figure 7.3.1).
- 22.⁺ Call a graph *irreducible* if it is not separated by any complete subgraph. Every finite graph G can be decomposed into irreducible induced subgraphs, as follows. If G has a separating complete subgraph S , then decompose G into proper induced subgraphs G' and G'' with $G = G' \cup G''$ and $G' \cap G'' = S$. Then decompose G' and G'' in the same way, and so on, until all the graphs obtained are irreducible. By Exercise 20, G has a simplicial tree-decomposition into these irreducible subgraphs. Show that they are uniquely determined if the complete separators were all chosen minimal.
23. If $\mathcal{F}$ is a family of sets, then the graph G on $\mathcal{F}$ with $XY \in E(G) \Leftrightarrow X \cap Y \neq \emptyset$ is called the *intersection graph* of $\mathcal{F}$. Show that a graph is chordal if and only if it is isomorphic to the intersection graph of a family of (vertex sets of) subtrees of a tree.
24. Show that for $n \geq 3$ the graphs K^n , C^n , an arbitrary tree of order n , and the $n \times n$ grid have tree-decompositions of widths $n - 1$, 2 , 1 , and n , respectively. For K^n and C^n show that these values are best possible.
25. Can the tree-width of a subdivision of a graph G be smaller than $\text{tw}(G)$? Can it be larger?
26. Show that the tree-width of a finite graph is at least its minimum degree. Is this still true for infinite graphs?
- 27.⁺ Show that if a graph has circumference $k \neq 0$, then its tree-width is at most $k - 1$.
- 28.⁺ A graph is called *outerplanar* if it has a drawing in which every vertex lies on the boundary of the outer face. Show that outerplanar graphs can have arbitrarily large tree-width, or find the best upper bound.

A tree-decomposition whose tree is a path is a *path-decomposition*. The *path-width* $\text{pw}(G)$ of G is the least width of a path-decomposition of G .

29. Show that a graph has a path-decomposition into complete graphs if and only if it is isomorphic to an interval graph. (Interval graphs are defined in Exercise 44, Ch. 5.)
30. (continued)
Prove the following analogue of Proposition 12.4.4 for path-width: every graph G satisfies $\text{pw}(G) = \min \omega(H) - 1$, where the minimum is taken over all interval graphs H containing G .
- 31.⁺ Do trees have unbounded path-width?

A *transaction* of a sequence $(v_1, \dots, v_n)$ of vertices is a set of disjoint paths from an initial segment $\{v_1, \dots, v_i\}$ to the rest, $\{v_{i+1}, \dots, v_n\}$.

- 32.⁺ Given $k \in \mathbb{N}$ and a sequence $v_1, \dots, v_n$ of vertices in a graph G , show that G has a path-decomposition $(V_1, \dots, V_n)$ of adhesion $\leq k$, with $v_i \in V_i$ for all i , if and only if G contains no transaction $\mathcal{P}$ of $(v_1, \dots, v_n)$ of order $|\mathcal{P}| > k$.
33. Show that the cycle C^n has connected tree-width $\lceil n/2 \rceil$.
34. Show that the $n \times n$ grid has tree-width n .
- 35.⁻ Let $\mathcal{B}$ be a maximum-order bramble in a graph G . Show that every minimum-width tree-decomposition of G has a unique part covering $\mathcal{B}$.
- 36.⁻ Let $\mathcal{P}$ be a minor-closed graph property. Show that strengthening the notion of a minor (for example, to that of topological minor) increases the set of forbidden minors required to characterize $\mathcal{P}$.
37. Deduce from the graph minor theorem that every minor-closed property can be expressed by forbidding finitely many topological minors. Is the same true for every property that is closed under taking topological minors?

Call a set $X \subseteq V(G)$ of vertices k -connected in G if $|X| \geq k$ and for all subsets $Y, Z \subseteq X$ with $|Y| = |Z| \leq k$ there are $|Y|$ disjoint Y - Z paths in G .

- 38.⁺ Show that the tree-width of a graph G is large if and only if it contains a large set of vertices that is k -connected in G for some large k . For example, show that graphs of tree-width $< k$ contain no $(k+1)$ -connected set of $3k$ vertices, and that graphs containing no $(k+1)$ -connected set of $3k$ vertices have tree-width $< 4k$.
39. (continued)
- (i)⁺ Find an $\mathbb{N} \rightarrow \mathbb{N}^2$ function $k \mapsto (h, \ell)$ such that every graph with an ℓ -connected set of h vertices contains a bramble of order $> k$.
- (ii)⁻ Using the last exercise, deduce the following weakening of the difficult implication of Theorem 12.4.3: given k , every graph of large enough tree-width $f(k)$ contains a bramble of order $> k$.
- 40.⁻ Show that if separations r, s are nested, they have orientations $\vec{r} \geq \vec{s}$.
41. Two cuts are *nested* if a side of one is contained in a side of the other. Show that the k -cuts in a k -edge-connected graph are nested if k is odd.
42. Characterize the 2-tangles in a graph.
43. Find a 4-tangle in the 3-dimensional cube.

When τ is a tangle and $(A, B) \in \tau$, we call A the *small side* of $\{A, B\}$ in τ and B its *big side*.

44. Let G be a graph with a tangle τ of order k .
- (i) Show that every graph $G' \succ G$ also has a k -tangle.
- (ii) Justify the notion of a ‘small side’ by showing that if $(A, B) \in \tau$ and $\{A', B'\}$ is a separation of order $< k$ with $A' \subseteq A$ or $B' \supseteq B$, then $(A', B') \in \tau$.

- (iii) Deduce from the profile property of τ that the intersection X of the big sides of any star in τ contains at least k vertices.
- (iv) Deduce that, for every set X of fewer than k vertices, exactly one of the components C of $G - X$ is ‘big’, in the sense that $(V(G - C), X \cup V(C)) \in \tau$.

45.⁺ Is (iv) of the previous exercise true also in infinite graphs (with k finite)?

A *decider set* for a tangle τ in G is a set $X \subseteq V(G)$ such that $|X \cap A| < |X \cap B|$ for every $(A, B) \in \tau$.

46.⁺ Show that a k -tangle induced by a $2k$ -tangle always has a decider set.

47. Show that separations $\{A, B\}$ and $\{C, D\}$ cross if and only if both A and B meet both C and D outside $A \cap B \cap C \cap D$.

48. Does every set of separations in a graph admit a consistent orientation?

49. Show that for every consistent orientation O of the set N of the separations $S_e := \{U_1, U_2\}$ in Lemma 12.3.1, one for every edge e of T , there exists a node t of T such that O orients every separation in N towards V_t .

50.⁻ Show that orienting the edges of a tree $T = (V, E)$ towards some fixed node t is consistent for the partial ordering on $\vec{E}$ defined in Section 12.5. Is this a bijection between V and the consistent orientations of E ?

51. (continued)

Show that S -trees are just a formal way to display, by a concrete tree, the ‘tree-like’ structure of nested sets of separations:

- (i) Given an S -tree (T, α) , show that $\alpha(\vec{E}(T)) \subseteq \vec{S}$ is nested.
(A set $\sigma \subseteq \vec{S}$ is *nested* if $\{s \mid \vec{s} \in \sigma\}$ is nested.)
- (ii)⁺ Given a nested set S of separations, construct an S -tree (T, α) with surjective $\alpha: \vec{E}(T) \rightarrow \vec{S}$.
- (iii) Derive Exercise 16 (ii).

52. (continued)

Prove the assertions about trees of tangles and tree-decompositions made in the long paragraph following the proof of Theorem 12.5.8.

53.⁺ Show that in any graph of order n there are at most n tangles.

54. Recall from the proof of Theorem 12.4.3 how a k -bramble orients S_k . Is this orientation of S_k a tangle? Is it a profile?

55. Show the following implications for a graph G :

- (i) G contains a k -block $\Rightarrow G$ has a bramble of order k .
- (ii) G has a tangle of order $k \Rightarrow G$ has a bramble of order k .
- (iii) G has a bramble of order $3k \Rightarrow G$ has a tangle of order k .
- (iv) G contains a k -block $\Rightarrow G$ has a k -tangle or $|B| \leq \frac{3}{2}(k-1)$.

Is there a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that, for every $k \in \mathbb{N}$, if G has a tangle of order at least $f(k)$ then it contains a k -block?

56. Show that k -blocks of size at least $3k/2$ define robust profiles of S_k . Strengthen Theorem 12.3.7 for such blocks, to find a canonical tree-decomposition which, for variable k , efficiently distinguishes all pairs of k -blocks of size at least $3k/2$ such that neither includes the other.
- 57.⁺ Define as the *order* of a bipartition $\{A, B\}$ of the vertex set of a graph G the number $\|A, B\|$ of its A – B edges. Define *orientations* of such bipartitions in the obvious way. Call an orientation of the set of all the vertex bipartitions of order $< k$ an *edge-tangle of order k* if it has no three elements $(A_1, B_1), (A_2, B_2), (A_3, B_3)$ such that $A_1 \cup A_2 \cup A_3 = V(G)$. Show that the proofs of Theorems 12.5.1 and 12.5.8 yield tree-of-tangle theorems for edge tangles, the second of them canonical.
58. Discuss the following three shortcuts in the proof of Theorem 12.5.11, all designed to avoid shifting. Does any of them work?
- (i) In the induction step, pick any separation $s \in S_k \setminus S_O$ of your choice. (If you think it helps, choose it of minimum order.) Add each of its two orientations $\vec{s}, \overleftarrow{s}$ to O to obtain S_k -trees that are over $\mathcal{T}^* \cup \mathcal{T}^+$ except at one leaf, whose label is $\{\overleftarrow{s}\}$ or $\{\vec{s}\}$, respectively. Merge the trees at these leaves as in the text.

Does it worry you that the extensions of O are no longer closed upwards in $\overrightarrow{S_k}$? If so, why? Then try the following adaptations of (i):

- (ii) Dispense with the requirement that O be closed upwards in $\overrightarrow{S_k}$.
 - (iii) Add to O not just $\vec{s}$ or $\overleftarrow{s}$, but their entire up-closure in $\overrightarrow{S_k}$.
59. To illustrate the equivalence of Theorems 12.4.3 and 12.5.11, prove the following assertions for all graphs G :
- (i) G has tree-width $< k - 1$ if and only if it has an S_k -tree over $\mathcal{F}_k$.
 - (ii) G has a k -bramble if and only if it admits an $\mathcal{F}_k$ -tangle of S_k .
- 60.⁺ Modify the proof of Theorem 12.5.9 to obtain a proof of Theorem 12.5.11.
61. Extend Theorem 12.6.5 as follows. Let H be a connected planar graph, let $\mathcal{X}$ be any set of connected graphs including H , and let $\mathcal{H} := \{IX \mid X \in \mathcal{X}\}$. Show that $\mathcal{H}$ has the Erdős-Pósa property, witnessed by the same function f as defined in the proof of Theorem 12.6.5. Explain how it is possible that f depends on H but not on any of the other graphs in $\mathcal{X}$.
- 62.⁺ Show that, for every non-planar graph H , the class IH fails to have the Erdős-Pósa property.
- (Hint. Embed H in a surface.)
- 63.⁺ Let $\mathcal{H}$ be a class of connected graphs, and $k \geq 1$ an integer. Without using any theorems from this chapter, show that $\mathcal{H}$ has the Erdős-Pósa property for graphs without a k -tangle.

- 64.⁺ Show that the four ingredients to the structure of the graphs in $\text{Forb}_{\preceq}(K^n)$ as described in Theorem 12.6.6 – tree-decomposition, an *apex set* X , arbitrary surfaces $S \not\preceq K^n$, and *vortices* $H_1, \dots, H_k$ – are all needed to capture all the graphs in $\text{Forb}_{\preceq}(K^n)$. More precisely, find examples of graphs in $\text{Forb}_{\preceq}(K^n)$ showing that Theorem 12.6.6 becomes false if we require in addition that the tree-decomposition has only one part, or that X is always empty, or that S is always the sphere, or that $H_1, \dots, H_k$ are always empty. No exact proofs are required.
- 65.⁺ (continued)
 Show that, unlike in Theorem 12.6.6, the surfaces used in Theorem 12.6.7 cannot be limited to those in which K^n cannot be drawn. (As before, no exact proofs are required.)
66. Without using the graph minor theorem, show that the chromatic number of the graphs in any $\preceq$ -antichain is bounded.
67. Let S_g denote the orientable surface obtained from the sphere by adding g handles. Find a lower bound for $|\mathcal{K}_{\mathcal{P}(S)}|$ in terms of g .
 (Hint. The smallest g such that a given graph can be embedded in S_g is its *orientable genus*. Use the theorem that the orientable genus of a graph is equal to the sum of the genera of its blocks.)
68. Deduce the graph minor theorem from the self-minor conjecture.
69. Prove Theorem 12.6.9, assuming that G has a normal spanning tree.
70. Let G be a locally finite graph obtained from the $\mathbb{Z} \times \mathbb{Z}$ grid H by adding an infinite set of edges xy with $d_H(x, y)$ unbounded. Show that $G \succ K^{\aleph_0}$. Can you do the same if the distances $d_H(x, y)$ are bounded (but at least 3)?
71. Is the infinite $\mathbb{Z} \times \mathbb{Z}$ grid a minor of the $\mathbb{Z} \times \mathbb{N}$ grid? Is the latter a minor of the $\mathbb{N} \times \mathbb{N}$ grid?
- 72.⁺ Extend Proposition 12.3.6 to infinite graphs not containing an infinite complete subgraph.
73. Using the previous exercise, prove that if every finite subgraph of G has tree-width less than $k \in \mathbb{N}$ then so does G .
74. Show that no assumption of large finite connectivity can ensure that a countable graph has a K^r minor when $r \geq 5$. However, using the previous exercise show that sufficiently large finite connectivity forces an infinite graph to contain any given planar minor.

Notes

Robertson & Seymour have traditionally referred to the graph minor theorem as *Wagner’s conjecture*. Wagner did indeed discuss this problem in the 1960s with his then students, Halin and Mader, and it seems that Mader conjectured a positive solution. Wagner himself always insisted that he did not – even after the graph minor theorem had been proved.

Robertson & Seymour’s proof of the graph minor theorem is given in the numbers IV–VII, IX–XII and XIV–XXII of their series of over 20 papers under the common title of *Graph Minors*, most of which appeared in the *Journal of Combinatorial Theory, Series B*, between 1983 and 2012. Of their theorems cited in this chapter, Theorem 12.4.2 is from Graph Minors IV, Theorems 12.5.9 and 12.5.1 from Graph Minors X, Theorems 12.6.3 and 12.6.5 from Graph Minors V, and Theorem 12.6.6 from Graph Minors XVI.

Kruskal’s theorem on the well-quasi-ordering of finite trees was first published in J.B. Kruskal, Well-quasi ordering, the tree theorem, and Vászonyi’s conjecture, *Trans. Amer. Math. Soc.* **95** (1960), 210–225. Our proof is due to Nash-Williams, who introduced the versatile proof technique of choosing a ‘minimal bad sequence’. This technique was also used in our proof of Higman’s Lemma 12.1.3.

Nash-Williams generalized Kruskal’s theorem to infinite graphs. This extension is much more difficult than the finite case. Its proof introduces as a tool the notion of *better-quasi-ordering*, a concept that has profoundly influenced well-quasi-ordering theory. The graph minor theorem is false for uncountable graphs; this was shown by R. Thomas, A counterexample to ‘Wagner’s conjecture’ for infinite graphs, *Math. Proc. Camb. Phil. Soc.* **103** (1988), 55–57. Whether or not the countable graphs are well-quasi-ordered as minors, and whether the finite (or the countable) graphs are better-quasi-ordered as minors, are related questions that remain wide open. Both are related also to the self-minor conjecture. This, too, was originally intended to include graphs of arbitrary cardinality, but was disproved for uncountable graphs by B. Oporowski, A counterexample to Seymour’s self-minor conjecture, *J. Graph Theory* **14** (1990), 521–524.

Doubling all the edges of an cycle, and then subdividing every new edge once, yields a graph which is incomparable under the topological minor relation to any graph obtained in this way from a cycle of any other length. This shows that the finite graphs are not well-quasi-ordered by the topological minor relation; cf. Exercise 10. C.-H. Liu, *Graph Structures and Well-Quasi-Ordering*, PhD thesis, Georgia Institute of Technology (2014), showed that this example is essentially the only obstruction to well-quasi-ordering by the topological minor relation: given any integer k , the finite graphs that do not contain, as a topological minor, a graph obtained from a path P^k in this way are well-quasi-ordered as topological minors.

The notions of tree-decomposition and tree-width were first introduced (under different names) by R. Halin, *S-functions for graphs*, *J. Geometry* **8** (1976), 171–186. Among other things, Halin showed that grids can have arbitrarily large tree-width. Robertson & Seymour reintroduced the two concepts, apparently unaware of Halin’s paper, with direct reference to K. Wagner, Über eine Eigenschaft der ebenen Komplexe, *Math. Ann.* **114** (1937), 570–